



From Chaos to Compliance With

MICROSOFT INTUNE REMEDIATIONS



WHO AM I?

Anthony Porter

Cloud Security Architect

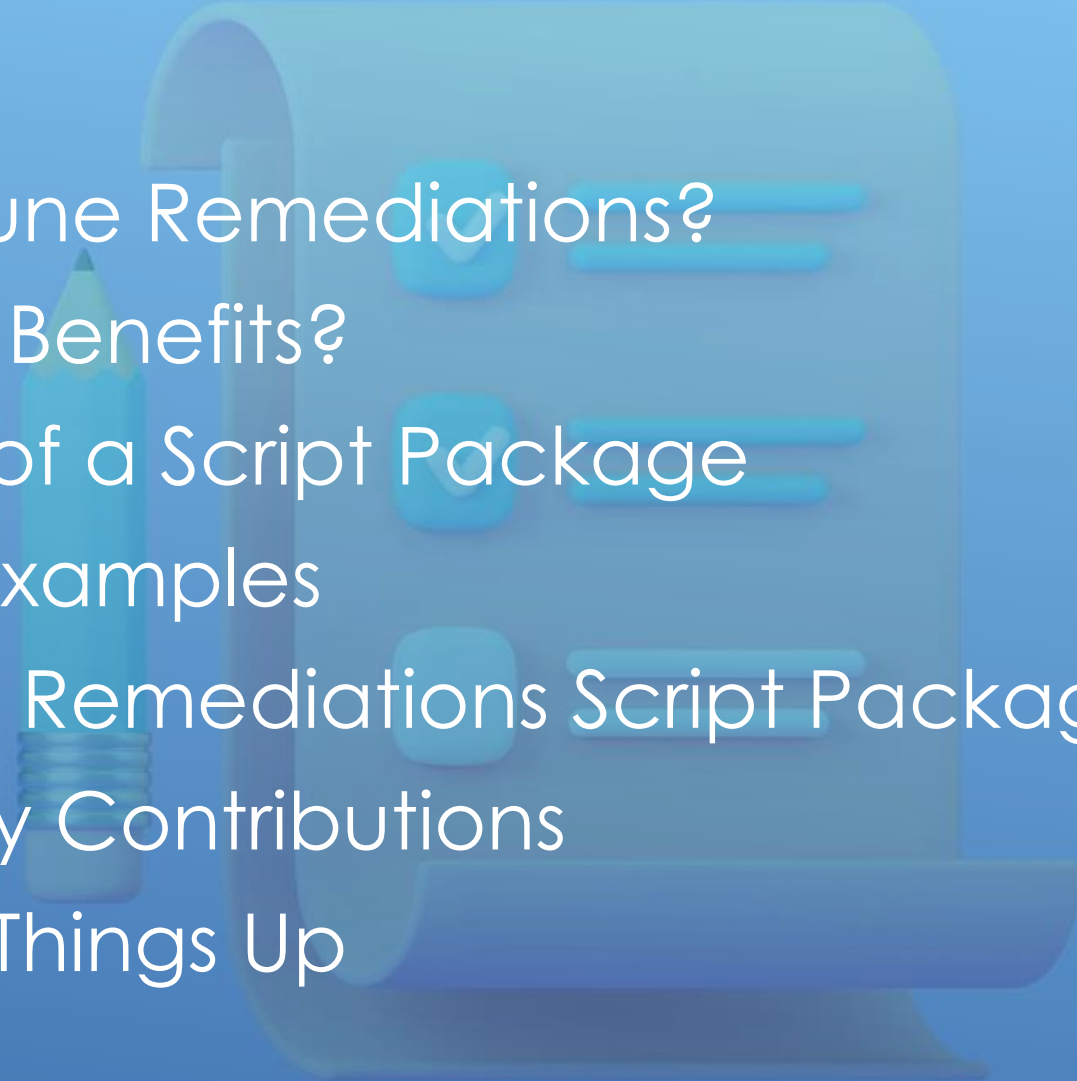
Canon Business Services ANZ

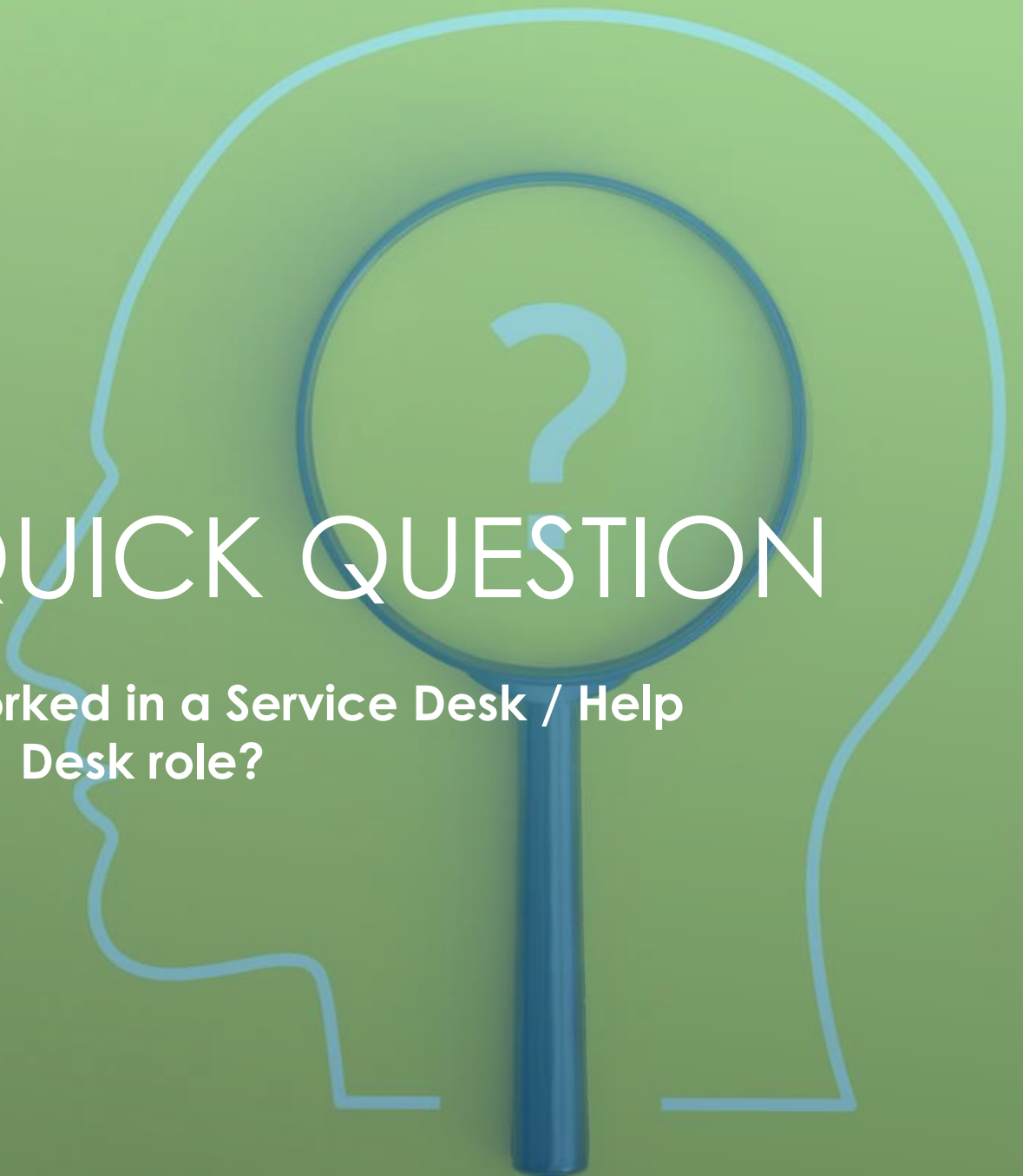
- Blog: <https://antoporter.github.io>
- LinkedIn: <https://www.linkedin.com/in/AnthonyAntoPorter>
- GitHub: <https://www.github.com/AntoPorter>

WHAT IS INTUNE REMEDIATIONS?



AGENDA

- ▶ What is Intune Remediations?
 - ▶ What's the Benefits?
 - ▶ Properties of a Script Package
 - ▶ Package Examples
 - ▶ Creating a Remediations Script Package
 - ▶ Community Contributions
 - ▶ Wrapping Things Up
- 
- A faint, stylized illustration in the background shows a light blue notepad with rounded corners and a pencil standing vertically to its left. The notepad has several horizontal lines, some of which are highlighted in a darker blue, suggesting a checklist or agenda. The entire scene is set against a solid blue gradient background.



JUST A QUICK QUESTION

Who here has worked in a Service Desk / Help
Desk role?

SO, WHAT'S THE BENEFITS?



BENEFITS TO INTUNE REMEDIATIONS



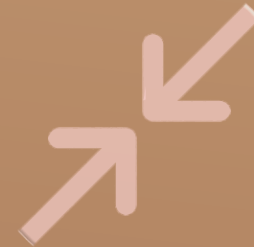
Customizable



Reportability



Scheduling /
On-Demand



Ease of Use



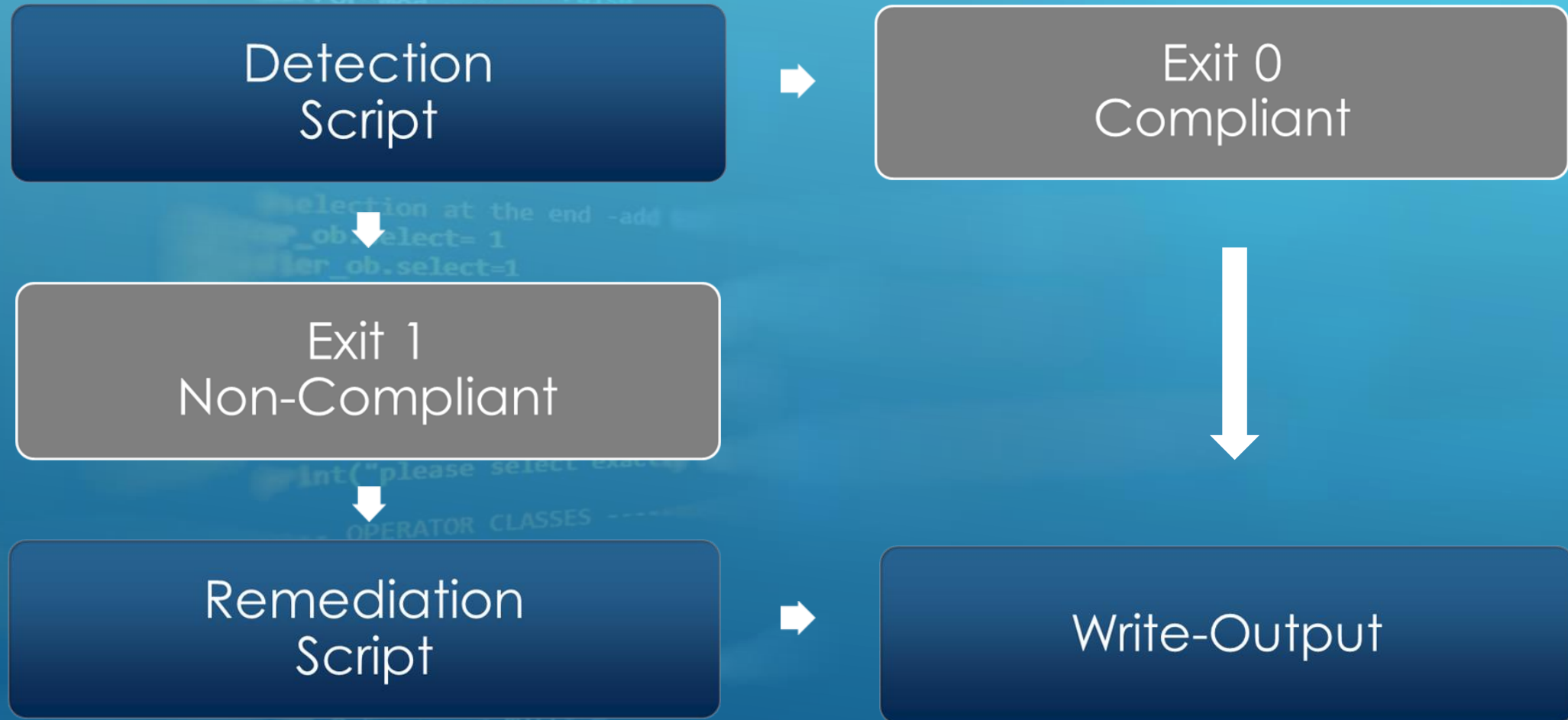
PowerShell
Capabilities

LET'S DISMANTLE A SCRIPT PACKAGE

```
object to mirror_
mirror_mod.mirror_object =
operation == "MIRROR_X":
mirror_mod.use_x = True
mirror_mod.use_y = False
mirror_mod.use_z = False
operation == "MIRROR_Y":
mirror_mod.use_x = False
mirror_mod.use_y = True
mirror_mod.use_z = False
operation == "MIRROR_Z":
mirror_mod.use_x = False
mirror_mod.use_y = False
mirror_mod.use_z = True
selection at the end -add
ob.select= 1
mirror_ob.select=1
context.scene.objects.active
("Selected" + str(modifier_
mirror_ob.select = 0
= bpy.context.selected_object
data.objects[one.name].select
print("please select exactly
-- OPERATOR CLASSES --
```

```
types.Operator):
X mirror to the selected
object.mirror_mirror_x"
mirror X"
```

PROPERTIES OF A SCRIPT PACKAGE



LET'S GET INTO SOME
EXAMPLES

DEVICE COMPLIANCE SCRIPTS

Confirm BitLocker Is Enabled

```
1 # Detection Script: Detect_BitLocker.ps1
2
3 # Check if BitLocker is enabled
4 $bitLockerStatus = Get-BitLockerVolume -MountPoint "C:"
5
6 if ($bitLockerStatus.ProtectionStatus -ne "On") {
7     Write-Output "BitLocker is not enabled on the system drive."
8     exit 1
9 } else {
10    Write-Output "BitLocker is enabled on the system drive."
11    exit 0
12 }
```

```
1 # Remediation Script: Remediate_BitLocker.ps1
2
3 # Enable BitLocker on the system drive
4 Enable-BitLocker -MountPoint "C:" -EncryptionMethod XtsAes256 -UsedSpaceOnly -TpmProtector
5
6 Write-Output "BitLocker has been enabled on the system drive."
```

Confirm Firewall Profiles are Enabled

```
1 # Detection Script: Detect_Firewall.ps1
2
3 # Check if the firewall is enabled
4 $firewallStatus = Get-NetFirewallProfile -Profile Domain,Public,Private
5
6 foreach ($profile in $firewallStatus) {
7     if ($profile.Enabled -eq $false) {
8         Write-Output "Firewall is disabled for profile: $($profile.Name)"
9         exit 1
10    }
11 }
12
13 Write-Output "Firewall is enabled for all profiles."
14 exit 0
```

```
# Remediation Script: Remediate_Firewall.ps1

# Enable the firewall for all profiles
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled True

Write-Output "Firewall has been enabled for all profiles."
```

DEVICE CONFIGURATION SCRIPTS

Mapping Network Drives

```
1 # Detection Script: Detect_DriveMapping.ps1
2
3 # Define the network drive letter and path
4 $driveLetter = "Z:"
5 $networkPath = "\\server\share"
6
7 # Check if the drive is mapped
8 $drive = Get-PSDrive -Name $driveLetter -ErrorAction SilentlyContinue
9
10 if ($null -eq $drive -or $drive.Root -ne $networkPath) {
11     Write-Output "Network drive not mapped: $driveLetter"
12     exit 1
13 } else {
14     Write-Output "Network drive is mapped: $driveLetter"
15     exit 0
16 }
```

```
# Remediation Script: Remediate_DriveMapping.ps1

# Define the network drive letter and path
$driveLetter = "Z:"
$networkPath = "\\server\share"

# Map the network drive
New-PSDrive -Name $driveLetter -PSProvider FileSystem -Root $networkPath -Persist

Write-Output "Network drive has been mapped: $driveLetter"
```

Installing Corporate Certificates

```
1 # Check if the certificate is installed
2 $cert = Get-ChildItem -Path Cert:\LocalMachine\My | Where-Object { $_.Subject -eq "CN=CorporateCert" }
3 if ($cert) {
4     Write-Output "Certificate is installed"
5     exit 0
6 } else {
7     Write-Output "Certificate is not installed"
8     exit 1
9 }
```

```
# Install the certificate
Import-Certificate -FilePath "C:\Path\To\CorporateCert.cer" -CertStoreLocation Cert:\LocalMachine\My
Write-Output "Certificate installed"
```

MISCELLANEOUS SCRIPT PACKAGE IDEAS

Defender AV – Real Time Protection

```
1
2  if((Get-MpComputerStatus).RealTimeProtectionEnabled -eq "True") {
3      Write-Output "Device Compliant"
4      exit 0
5  } else {
6      Write-Output "Device Non-Compliant"
7      exit 1
8  }
```

```
1  try {
2      Set-MpPreference -DisableRealtimeMonitoring $false
3      Write-Output "Device Remediated"
4      exit 0
5  }
6  catch {
7      Write-Output "Remediation Failed"
8      exit 1
9  }
```

Custom Toast Notifications

```
1  # Detection script
2  exit 1
```

```
1  # Define the toast notification content
2  # Group defines the notification grouping (eg. category). This allows for multiple use cases, which all align within the same group.
3  # Title defines the heading of the Toast Notification.
4  # Message defines the contents of the Toast Notification
5
6  $Group = "This is a Notification!"
7  $Title = "This is the Title!"
8  $Message = "This is the Message!"
9
10 # Create the toast notification
11 [Windows.UI.Notifications.ToastNotificationManager, Windows.UI.Notifications, ContentType = WindowsRuntime] | Out-Null
12 $template = [Windows.UI.Notifications.ToastTemplateType]::ToastText02
13 $toastXml = [Windows.UI.Notifications.ToastNotificationManager]::GetTemplateContent($template)
14 $toastTextElements = $toastXml.GetElementsByTagName("text")
15 $toastTextElements.Item(0).AppendChild($toastXml.CreateTextNode($Title)) | Out-Null
16 $toastTextElements.Item(1).AppendChild($toastXml.CreateTextNode($Message)) | Out-Null
17 $toast = [Windows.UI.Notifications.ToastNotification]::new($toastXml)
18 $notifier = [Windows.UI.Notifications.ToastNotificationManager]::CreateToastNotifier($Group)
19 $notifier.Show($toast)
```

Hey You!

Yeah You!
I'm Watchin You!

REPORTING SCRIPT PACKAGE IDEAS

Certificate Expiry Report

```
1 # Check for certificates nearing expiry
2 $certificates = Get-ChildItem -Path Cert:\LocalMachine\My | Where-Object { $_.NotAfter -lt (Get-Date).AddDays(30) } | Select-Object Subject, NotAfter
3
4 # Output the certificates nearing expiry
5 # Write-Output $certificates
6
7 $csvPath = "C:\temp\CertificateExpiryStatus.csv"
8
9 $certificates | Export-Csv -Path $csvPath -NoTypeInformation
10 Write-Output "Certificate Expiry status exported to $csvPath"
11
12 Exit 0
```

Disk Space Usage Report

```
1 # Check disk space usage
2 $diskSpace = Get-PSDrive -PSProvider FileSystem | Select-Object Name, @{Name="Used(GB)";Expression={[math]::round($_.Used/1GB,2)}}, @{Name="Free(GB)";Expression={[math]::round($_.Free/1GB,2)}}
3
4 # Output the disk space usage
5 # Write-Output $diskSpace
6
7 $csvPath = "C:\temp\DiskSpaceStatus.csv"
8
9 $diskSpace | Export-Csv -Path $csvPath -NoTypeInformation
10 Write-Output "Disk Space status exported to $csvPath"
11
12 Exit 0
```

Pending Reboot Report

```
1 # Check for pending reboot
2 $pendingReboot = Get-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Component Based Servicing\RebootPending" -ErrorAction SilentlyContinue
3
4 if ($pendingReboot) {
5     Write-Output "Reboot is pending."
6 } else {
7     Write-Output "No reboot pending."
8 }
9
10 Exit 0
```

Event Log Error Report

```
1 # Check for errors in the event log
2 $eventErrors = Get-EventLog -LogName System -EntryType Error -Newest 100 | Select-Object TimeGenerated, Source, EventID, Message
3
4 # Output the event log errors
5 # Write-Output $eventErrors
6
7 $csvPath = "C:\temp\EventLogErrorStatus.csv"
8
9 $eventErrors | Export-Csv -Path $csvPath -NoTypeInformation
10 Write-Output "Event Log Error status exported to $csvPath"
11
12 Exit 0
```

A close-up photograph of a cat's face, heavily tinted with a blue color. The cat has bright yellow eyes and its mouth is open, revealing its pink tongue and white teeth. The background is a soft, out-of-focus blue.

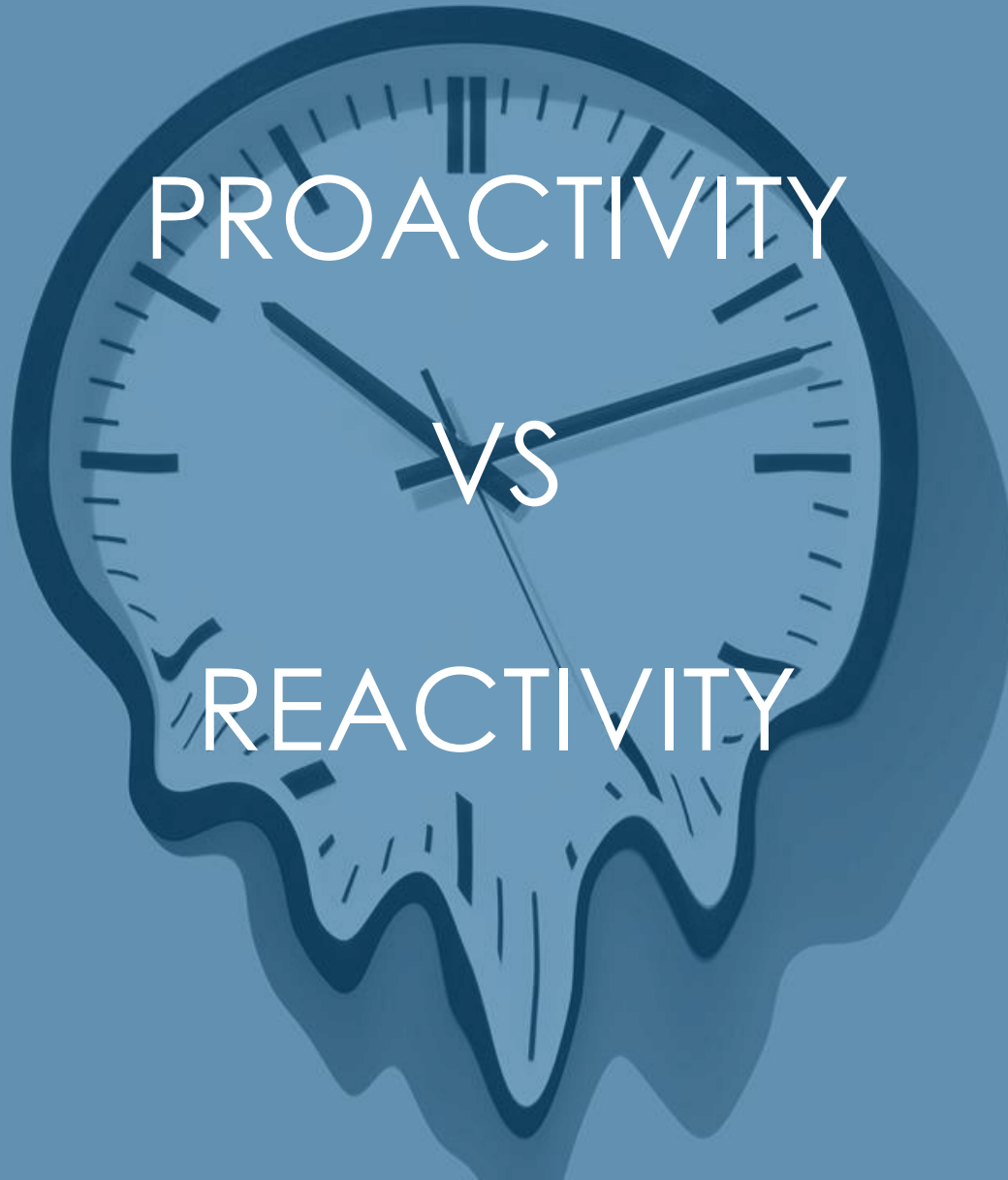
THE POSSIBILITIES ARE
ENDLESS*!

*subject to Limitations

PROACTIVITY

VS

REACTIVITY



PROACTIVE SCHEDULED

Examples:

- Configuration Checks
- Custom Compliance Requirements in your SOE
- Endpoint Metric Monitoring

REACTIVE ON-DEMAND

Examples:

- Start / Stop Services
- Clear Folder Contents
- Commence AV Scans

REPORTING



Devices > Remediation Script Package

Monitor > Device Status

REPORTING

- ▶ Pre-Remediation Detection Output
- ▶ Post-Remediation Detection Output

Search by device name		Detection status: 0 selected		Remediation status: 0 selected	
Device name	Detection status	Remediation status	OS version	Pre-remediation detection output	Post-remediation detection out...
SAT-L-PF2CL9DM	✔ Without issues	● Not run	10.0.19045.4780	RegKey Not Found, Complia... Review	
SAT-L-PF4HE0QH	✔ Without issues	● Not run	10.0.22621.3880	RegKey Not Found, Complia... Review	

- You can have up to 200 script packages.
- A script package can contain a detection script only or both a detection script and a remediation script.
 - A remediation script only runs if the detection script uses exit code `exit 1`, meaning the issue was detected.
- Ensure the scripts are encoded in UTF-8.
 - If the option **Enforce script signature check** is enabled in the **Settings** page of creating a script package, then make sure that the scripts are encoded in UTF-8 not UTF-8 BOM.
- The maximum allowed output size limit is 2048 characters.
- If the option **Enforce script signature check** is enabled in the **Settings** page of creating a script package, the script

SO, THAT LIMITATION

The maximum allowed output size limit is 2048 characters.

- Scripts built into Remediations are signed and the certificate is added to the **Trusted Publishers** certificate store of the device.
- When using third-party scripts that are signed, make sure the certificate is in the **Trusted Publishers** certificate store. As with any certificate, the certificate authority must be trusted by the device.
- Scripts without **Enforce script signature check** use the **Bypass** execution policy.
- Don't put reboot commands in detection or remediations scripts.
- Do not include any type of sensitive information in scripts (such as passwords)
- Do not include Personally Identifiable Information (PII) in scripts

A background image of a brick wall. The bricks are a mix of light blue and orange-brown colors, with some darker blue and orange streaks. The text is overlaid on the left side of the wall.

WONDERING HOW TO GET
STARTED?

Create custom script

✓ Basics **2 Settings** 3 Scope tags 4 Assignments 5 Review + create

i This script will run in detect-only mode because there is no remediation script.

Create a custom script package from scripts you've written. By default, scripts will run on assigned devices.

Detection script file *

Detection script

Remediation script file

Remediation script

Run this script using the logged-on credentials ☐ ☒

Enforce script signature check ☐ ☒

Run script in 64-bit PowerShell ☐ ☒

- ▶ Download/Create a Detection and Remediation Scripts
- ▶ Navigate to the Intune Portal
- ▶ Navigate to Devices > Scripts and remediations.
- ▶ Create a Script Package
- ▶ Upload the Detect/Remediation Scripts
- ▶ Configure Script Properties
- ▶ Set the Schedule
- ▶ Assign to Devices

CREATING A CUSTOM SCRIPT PACKAGE

Run this script using the logged-on credentials

Allows you to interact with user-specific data and settings

Enforce Script Signature Check

Enforcing a Script Signature Check ensures that only trusted signed publishers are executed

Run Script in 64-Bit PowerShell

Ensures scripts leverage the full capabilities of 64-Bit Environments, improving performance and compatibility with modern applications and systems

NOW THOSE PROPERTIES

Run this script using the logged-on
credentials

Yes

No

Enforce script signature check

Yes

No

Run script in 64-bit PowerShell

Yes

No

SCHEDULED REMEDIATIONS

Recurrences:

- Once
- Hourly
- Daily

1 Assignments **2 Review + save**

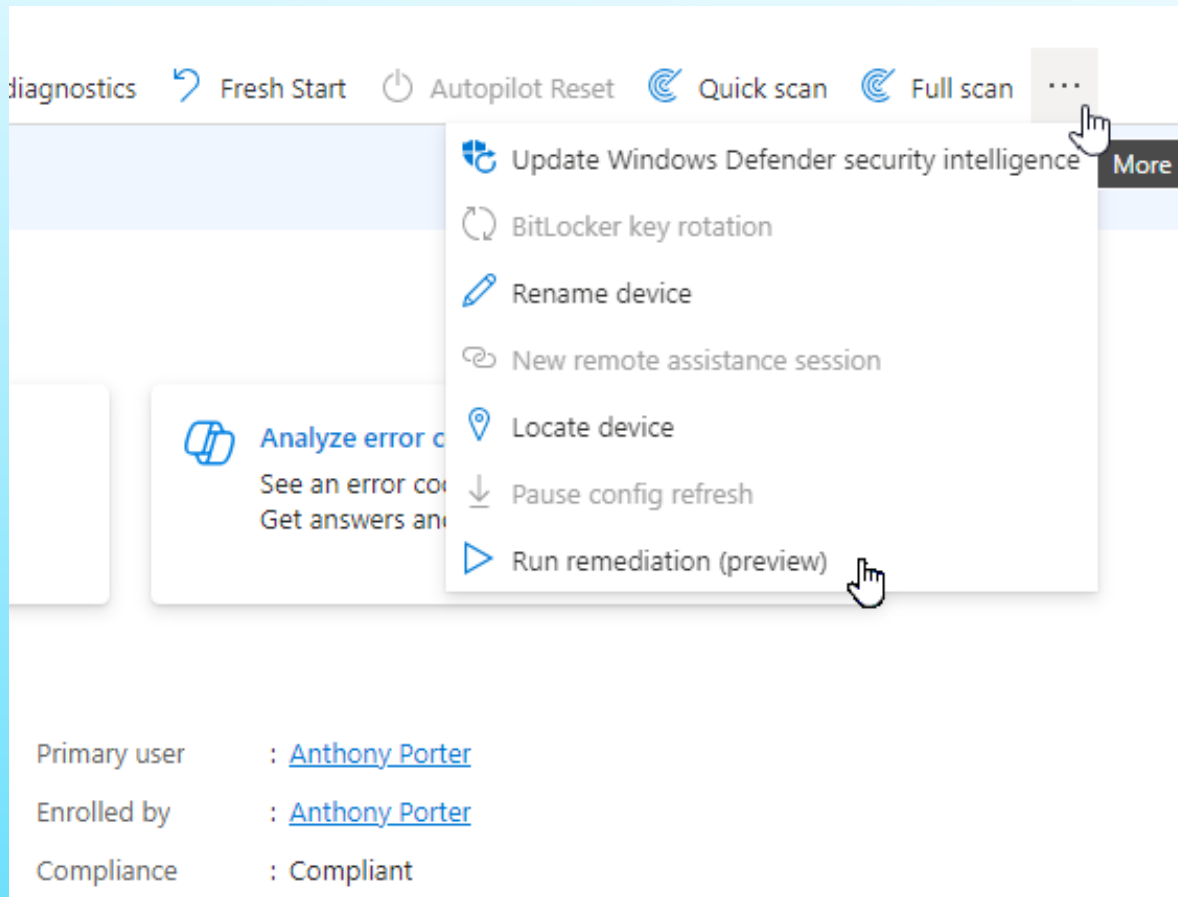
Select one or more groups to assign the script package.

Included groups

Assign to

Selected groups	Schedule	Filter	Filter mode
Intune-EndpointSec...	Daily	None	None

◀ ▶



ON-DEMAND REMEDiations

- Devices > Windows Endpoint
- Ellipsis (...) > Run Remediations (Preview)

AND THE REQUIREMENTS



Licensing Requirements

Windows 10/11 Enterprise E3 / E5

Windows 10/11 Education A3 / A5

Or Windows 10/11 Virtual Desktop Access (VDA) per user.



Device must be Entra Joined / Hybrid Joined



Device must be Intune managed

Running Windows 10 Enterprise/Pro/Education or Later

A glowing lightbulb hangs from a cord at the top center of the frame. The background is a deep blue, filled with numerous vertical light streaks and soft, out-of-focus circular bokeh lights, creating a dreamy, ethereal atmosphere. The lightbulb itself is illuminated from within, casting a warm, yellowish glow that contrasts with the cool blue tones of the background.

NEED SOME INSPIRATION?

main

1 Branch

0 Tags

Go to file

<> Code

 JayRHa Create LICENSE	85addde · 2 months ago	 143 Commits
 .images	update read me	last year
 0 - Template	Rename Uninstall-Silverlight to Detect-Silverlight	4 months ago
 Activate-Numlock	Add description	last year
 Add-Winget-App	Update remediate-app.ps1	last year
 AutomaticTimezone	added readme	5 months ago
 BlockAADWorkplaceJoin	BlockAADWorkplaceJoin	8 months ago
 Change-MultipleRegistryKeys	New script Change-MultipleRegistryKeys	10 months ago
 Change-Registry-Key-Generic	Add description	last year
 Check-DiskHealth	Add description	last year
 Check-PNPDevices	New script Check-PNPDevices	10 months ago
 Clear-DnsCache	Fix name	last year
 Clear-DownloadFolder-SingleUser	Rename Clear-DownloadFolderRemedaiton.ps1 to Clear-Do...	last year
 Clear-DownloadFolder	Rename Clear-DownloadFolderRemedaiton.ps1 to Clear-Do...	last year
 Clear-OutlookCache	Add description	last year
 Clear-TeamsCache	Add description	last year
 Create-LocalAdmin	Update Create-LocalAdminLAPSRemediation.ps1	10 months ago
 Detect-SCCM	Create Detect.ps1	3 months ago

About

No description, website, or topics provided.

-  Readme
-  MIT license
-  Activity
-  331 stars
-  41 watching
-  74 forks

Report repository

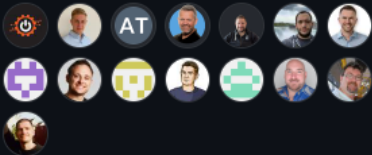
Releases

No releases published

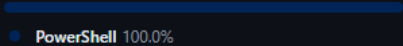
Packages

No packages published

Contributors 15



Languages



main 1 Branch 0 Tags

Go to file

Code

 aaronparker	Update Detect-AdobeAcrobatSettings.ps1	9e0659e · 3 months ago	61 Commits
.devops/pipelines	Update test-scripts.yml	last year	
.github	Merge pull request #3 from aaronparker/dependabot/github...	8 months ago	
.vscode	Update code coverage paths and upload step	last year	
AdobeAcrobat-Computer	Update Detect-AdobeAcrobatSettings.ps1	3 months ago	
AdobeAcrobat-Reader	Change keywords to lower case	last year	
AdobeAcrobat-User	Change keywords to lower case	last year	
Antivirus-Report	Move scripts back to root	last year	
AppX-Remove	Move scripts back to root	last year	
Device-DisableCoInstallers	Add Device-DisableCoInstallers	6 months ago	
DisableDomainCreds	Add DisableDomainCreds	7 months ago	
Feature-InternetExplorer	Remove Internet Explorer	9 months ago	
Groups-Administrators	Get-DeviceHealthScripts	last year	
LocalSecurityAuthority	Replace return with exit	7 months ago	
Locale	Change keywords to lower case	last year	
Location-Enable	Replace return with exit	7 months ago	
Microsoft365Apps-Update	Update Detect-Microsoft365AppsUpdate.ps1	last year	
MicrosoftTeams-Firewall	Replace return with exit	7 months ago	
MicrosoftTeams-GPU	Teams GPU	9 months ago	

About

PowerShell scripts for use with Remediations in Microsoft Intune

learn.microsoft.com/en-us/mem/analytic...

- windows powershell intune
- remediations proactive-remediations

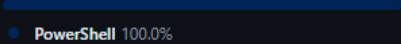
- Readme
- MIT license
- Activity
- 29 stars
- 5 watching
- 9 forks

Report repository

Contributors 2

-  **aaronparker** Aaron Parker
-  **dependabot[bot]**

Languages



main

1 Branch

0 Tags

Go to file

<> Code

 AntoPorter	remediations v3	ec84603 · 4 days ago	🕒 17 Commits
📁 DeviceCompliance	remediations-v2		2 months ago
📁 DeviceConfiguration	remediationupdates		last month
📁 DevicePerformance	remediations-v2		2 months ago
📁 MicrosoftDefenderAV	remediations-v2		2 months ago
📁 Miscellaneous	customscript		last month
📁 Reporting	remediations v3		4 days ago
📄 README.md	remediations v3		4 days ago

📖 README

Intune Remediations

The **Intune Remediations** collection is a set of script packages designed to detect and fix common support issues on user endpoints. Each script package includes a detection script, a remediation script, and metadata. By deploying these packages through Intune, you can proactively address issues before end-users even notice them, potentially reducing support calls.

Step-by-Step Guide: Using Microsoft Intune Remediations

About

I hope these Microsoft Intune Remediations inspire you to make your own!

script

intune

remediations

📖 Readme

📈 Activity

☆ 55 stars

👁 6 watching

🍴 11 forks

Report repository

Releases

No releases published

Packages

No packages published

Languages

PowerShell 100.0%

LET'S WRAP THINGS UP

Any Questions?

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Cloud Security Architect
Canon Business Services ANZ

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